

CASEMENT, Ireland

WMO: 039670

Lat: 53.300N

Lon: 6.433W

Elev: 97

StdP: 100.16

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.0	-1.5	-4.8	2.5	-0.9	-3.5	2.9	0.1	16.5	8.3	15.2	8.3	2.3	230	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	22.9	17.5	21.2	16.6	20.0	16.0	18.3	21.4	17.4	20.2	16.7	19.3	4.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.4	19.8	16.3	11.7	18.9	15.5	11.1	18.1	52.1	21.4	49.4	20.2	47.2	19.3	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.2	10.9	DB	-6.1	25.4	2.3	1.8	-7.8	26.7	-9.1	27.8	-10.4	28.8	-12.1	30.1	(4)
(5)			WB	-6.4	20.0	2.3	1.1	-8.1	20.8	-9.4	21.4	-10.7	22.0	-12.3	22.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	5.4	5.4	6.5	8.3	11.0	13.7	15.5	15.2	13.3	10.6	7.4	5.7	(6)
(7)		DBStd	4.60	2.98	2.84	2.75	2.75	2.63	2.37	2.18	2.10	2.31	2.74	3.05	3.41	(7)
(8)		HDD10.0	723	145	131	112	64	19	1	0	0	2	26	86	137	(8)
(9)		HDD18.3	3104	402	362	367	301	228	141	91	99	153	241	327	393	(9)
(10)		CDD10.0	670	1	2	4	13	49	111	171	162	99	44	9	3	(10)
(11)		CDD18.3	8	0	0	0	0	0	1	4	3	1	0	0	0	(11)
(12)		CDH23.3	34	0	0	0	0	1	6	19	8	1	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	0	1	1	0	0	0	0	(13)
(14)	Wind	WSAvg	5.1	6.2	5.9	5.6	4.8	4.7	4.4	4.2	4.4	5.2	5.4	5.9	(14)	
(15)	Precipitation	PrecAvg	740	67	50	49	51	59	56	52	64	77	77	74	(15)	
(16)		PrecMax	1021	122	128	99	124	149	179	152	172	157	179	184	206	(16)
(17)		PrecMin	573	10	8	7	5	15	6	9	3	7	14	12	22	(17)
(18)		PrecStd	103	30	30	21	26	29	36	28	34	35	41	40	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.6	15.3	18.7	21.3	24.1	25.4	22.0	18.2	15.3	13.7	(19)	
(20)			MCWB	11.6	11.2	11.3	13.0	15.7	16.6	18.4	18.6	17.5	15.6	13.7	12.5	(20)
(21)		2%	DB	11.7	12.0	13.3	15.9	19.2	21.5	23.2	21.9	20.0	16.7	14.0	12.6	(21)
(22)			MCWB	10.6	10.3	10.5	11.7	14.4	16.1	17.8	17.5	16.5	14.5	12.6	11.4	(22)
(23)		5%	DB	10.9	11.0	12.1	14.2	17.3	19.9	21.3	20.3	18.6	15.7	13.0	11.7	(23)
(24)			MCWB	9.8	9.7	10.0	10.8	13.3	15.3	16.8	16.6	15.5	13.8	11.7	10.5	(24)
(25)		10%	DB	10.0	9.9	11.0	12.9	15.6	18.4	19.8	19.2	17.3	14.9	12.0	10.7	(25)
(26)			MCWB	8.9	8.6	9.1	10.0	12.3	14.6	16.0	15.9	14.6	13.1	10.8	9.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.7	12.1	13.8	16.8	18.2	19.5	18.6	16.1	14.2	12.6	(27)	
(28)			MCDWB	12.5	12.8	13.7	17.3	20.9	22.0	23.6	22.9	20.8	17.5	15.0	13.4	(28)
(29)		2%	WB	10.8	10.7	11.1	12.4	15.0	16.8	18.4	18.2	17.1	15.0	12.9	11.6	(29)
(30)			MCDWB	11.6	11.8	12.7	15.1	18.2	20.1	21.9	20.7	19.2	16.3	13.8	12.4	(30)
(31)		5%	WB	9.9	9.7	10.3	11.3	13.8	15.9	17.5	17.3	16.1	14.1	11.9	10.6	(31)
(32)			MCDWB	10.8	10.8	11.9	13.7	16.6	19.2	20.5	19.7	18.1	15.5	12.8	11.5	(32)
(33)		10%	WB	9.0	8.8	9.3	10.3	12.8	15.0	16.6	16.5	15.1	13.3	10.9	9.7	(33)
(34)			MCDWB	9.9	9.9	10.9	12.5	15.2	17.8	19.2	18.8	16.8	14.7	12.0	10.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	5.7	6.7	7.4	7.7	7.8	7.6	7.1	6.2	5.4	5.2	(35)	
(36)			MCDBR	5.6	5.8	7.5	8.8	9.9	10.0	9.6	8.5	8.4	6.3	5.3	5.6	(36)
(37)			MCWBR	5.5	5.2	5.5	5.8	6.1	6.0	5.4	5.2	5.6	5.0	5.0	5.4	(37)
(38)		5% WB	MCDWB	5.7	5.8	6.8	8.1	8.8	8.9	8.6	7.8	7.5	5.9	5.2	5.7	(38)
(39)			MCWBR	5.7	5.4	5.5	5.7	5.8	5.8	5.5	5.2	5.6	5.2	5.1	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.345	0.372	0.387	0.393	0.382	0.385	0.366	0.359	0.347	0.327	0.323	(40)	
(41)		taud	2.258	2.327	2.277	2.268	2.293	2.355	2.351	2.417	2.433	2.415	2.329	2.231	(41)	
(42)		Ebn at Noon	616	748	804	842	855	868	857	856	815	740	627	546	(42)	
(43)		Edh at Noon	64	81	104	119	122	116	115	102	89	74	59	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.70	1.40	2.49	3.83	4.85	5.05	4.59	3.86	2.88	1.69	0.89	0.54	(44)	
(45)		RadStd	0.07	0.12	0.26	0.38	0.35	0.51	0.42	0.34	0.16	0.14	0.09	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page